2811(router) ani 2960(switch)

Exp2

ipconfig

ipconfig /all

ping google.com

tracert google.com

nslookup google.com

netstat

netstat -a

netstat -n

netstat -r

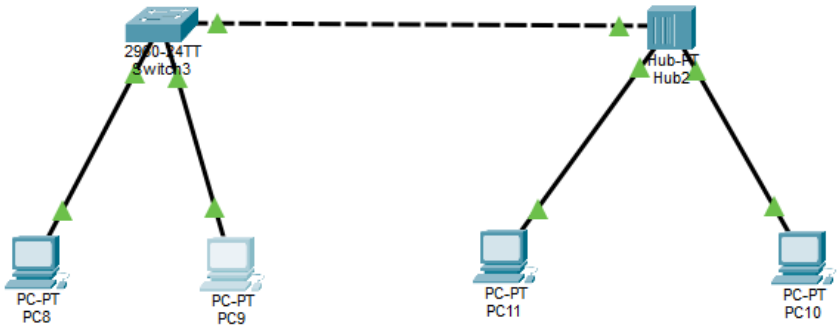
arp -a

hostname

getmac

pathping google.com

Exp3



Assign IP addresses manually.

Device	**IP Address**	**Subnet Mask**	enable configure terminal interface vlan 1 ip address 192.168.1.10 255.255.255.0 no shutdown end write memory
PC0	192.168.1.1	255.255.255.0	
PC1	192.168.1.2	255.255.255.0	
PC2	192.168.1.3	255.255.255.0	
PC3	192.168.1.4	255.255.255.0	

Exp4

 Add 2 Cisco 2811 routers, 2 Cisco 2960 switches, and 4 PCs in Packet Tracer.

 Connect the devices using cables: PCs → switches → routers, then connect both routers.

 Configure PC IPs: LAN 1 uses 192.168.1.2, 192.168.1.3; LAN 2 uses 192.168.2.2, 192.168.2.3. Gateways are 192.168.1.1 and 192.168.2.1.

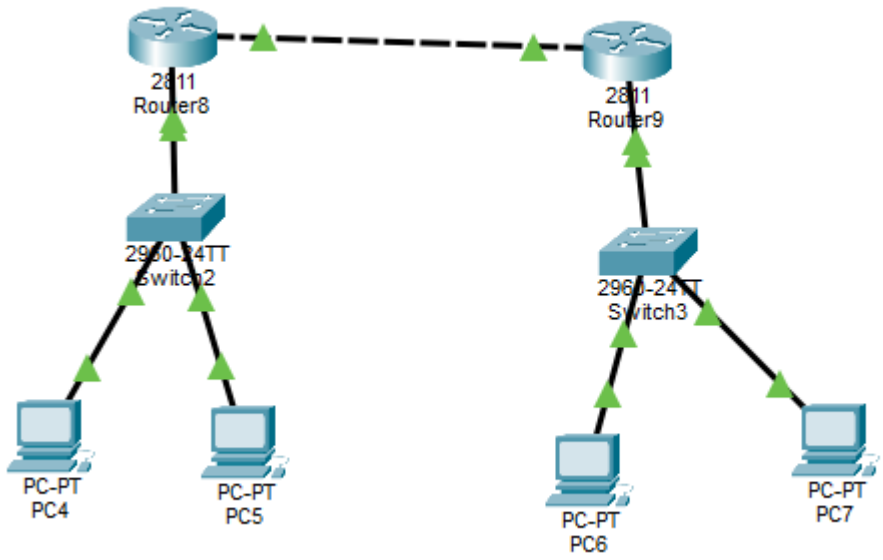
 Configure Router 1 interfaces: 192.168.1.1/24 toward LAN 1 and 192.168.3.1/24 toward Router 2.

 Configure Router 2 interfaces: 192.168.2.1/24 toward LAN 2 and 192.168.3.2/24 toward Router 1. Turn all interfaces On.

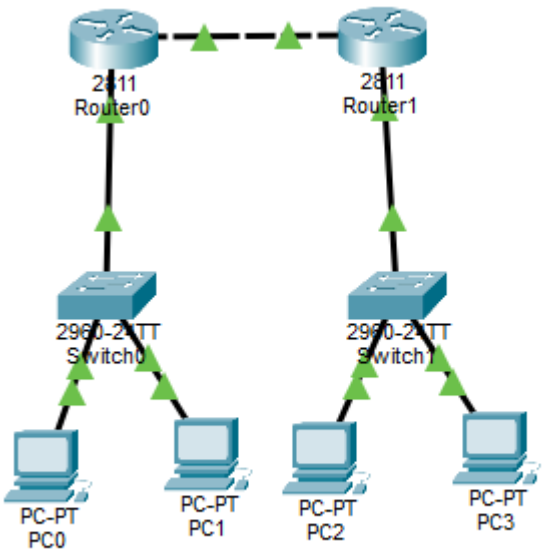
 On Router 1, go to Config → Routing → Static and add network 192.168.2.0, mask 255.255.255.0, next hop 192.168.3.2.

 On Router 2, add network 192.168.1.0, mask 255.255.255.0, next hop 192.168.3.1.

 Test communication between PCs using Add Simple PDU or ping. If the first attempt fails, try again after the links initialize.



Exp5



On Router0, open Config → RIP and add the network addresses connected to it, such as 192.168.1.0 and 192.168.2.0, using Add.

Exp6

```
import socket
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind(("localhost", 5000))
server.listen(1)
print("Server is waiting for connection...")
conn, addr = server.accept()
print("Connected by:", addr)
message = conn.recv(1024).decode()
print("Client:", message)
conn.send("Hello Client!".encode())
conn.close()
server.close()
```

```
import socket
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(("localhost", 5000))
```

```
client.send("Hello Server!".encode())
reply = client.recv(1024).decode()
print("Server:", reply)
client.close()
```

Exp7

```
import socket, struct
s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
s.bind(("localhost", 5001))
name, a = s.recvfrom(65535)
with open("received_" + name.decode(), "wb") as f:
    expected = 0
    while True:
        data, a = s.recvfrom(65535)
        n = struct.unpack("!I", data[:4])[0]
        if n == 0xFFFFFFFF:
            break
        if n == expected:
            f.write(data[4:])
            expected += 1
print("File received!")
s.close()
```

```
import socket, struct, os
s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
ip, port = "localhost", 5001
name = input("Enter file path: ")
s.sendto(os.path.basename(name).encode(), (ip, port))
with open(name, "rb") as f:
    n = 0
    while chunk := f.read(1024):
```

```
        s.sendto(struct.pack("!I", n) + chunk, (ip, port))
        n += 1
s.sendto(struct.pack("!I", 0xFFFFFFFF), (ip, port))
print("File sent!")
s.close()
```

Exp8

Exp9

```
enable
configure terminal
hostname R1
enable secret cisco123
no ip domain-lookup
service password-encryption
banner motd #Authorized Access Only!#

line console 0
password console123
login
exec-timeout 5 30
exit

line vty 0 4
password vty123
login
exec-timeout 5 30
exit

end
```

show running-config

copy running-config startup-config

show startup-config